Green's Theorem

Lecture 05 and 06

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Green's Theorem in the Plane

Circulation-Curl or Tangential Form

1. Let C be a piecewise smooth, simple closed curve enclosing R in the plane.
2. Let $\mathbf{F}(x, y) = M(x, y)\mathbf{i} + N(x, y)\mathbf{j}$ be a vector field with M and N having continuous first partial derivatives in an open region containing R.

Then the counterclockwise circulation of $\mathbf{F}$ around C equals the double integral of *curl* $\mathbf{F} \cdot \mathbf{k}$ over R.

$$\oint_C \mathbf{F} \cdot \mathbf{T}\, ds = \oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dxdy$$

The counterclocwise circulation of $\mathbf{F}$ around C defined by the line integral on the left hand side, is the integral of its rate of change (circulation density) over the region R enclosed by C, which is a double integral on the right hand side.

Flux-Divergence or Normal Form

1. Let C be a piecewise smooth, simple closed curve enclosing R in the plane.
2. Let $\mathbf{F}(x, y) = M(x, y)\mathbf{i} + N(x, y)\mathbf{j}$ be a vector field with M and N having continuous first partial derivatives in an open region containing R.

Then the outward flux of $\mathbf{F}$ across C equals the double integral of $div\ \mathbf{F}$ over R enclosed by C

$$\boxed{\oint_C \mathbf{F} \cdot \mathbf{n}\ ds = \oint_C M dy - N dx = \iint_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dxdy}$$

The outward flux of $\mathbf{F}$ across C defined by the line integral on the left hand side, is the integral of its rate of change (flux density) over the region R enclosed by C, which is a double integral on the right hand side.

(a) Both forms of Green's Theorem are equivalent.

(b) Both forms of Green's Theorem can be viewed as two-dimensional generalizations of the Net Change Theorem stated below:

Net Change Theorem (Fundamental Theorem of Calculus - Part II)

The net change in a differentiable function $F(x)$ over an interval $a \leq x \leq b$ is the integral of its rate of change:

$$\int_a^b F'(x)dx = F(b) - F(a)$$

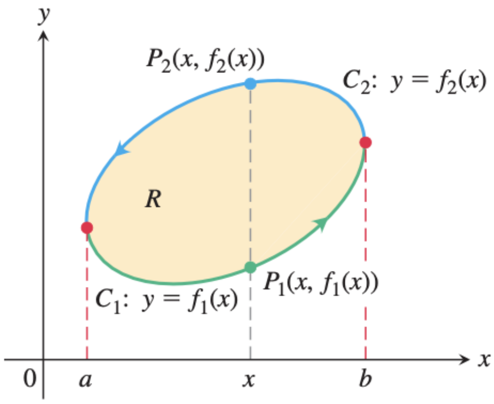
Proof of Green's Theorem for Special Regions

Context: Let C be a smooth simple closed curve in the $xy-$plane with the property that lines parallel to the axes cut it at no more than two points. Let R be the region enclosed by C and suppose that M, N, and their first partial derivatives are continuous at every point of some open region containing C and R.

To prove: We want to prove the circulation-curl form of Green's Theorem

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx dy$$

Consider the two functions with the following attributes



Here C is made up of two directed paths:

$$C_1 :\rightarrow y = f_1(x), \quad a \leq x \leq b$$
$$C_2 :\rightarrow y = f_2(x), \quad a \leq x \leq b$$

Mathematical Treatment: For any x between a and b, we can integrate $\dfrac{\partial M}{\partial y}$ with respect to y from $y = f_1(x)$ to $y = f_2(x)$ and obtain

$$\begin{aligned}\int_{f_1(x)}^{f_2(x)} \frac{\partial M}{\partial y} dy &= M(x, y)\Big|_{f_1(x)}^{f_2(x)} \\ &= M(x, f_2(x)) - M(x, f_1(x))\end{aligned}$$

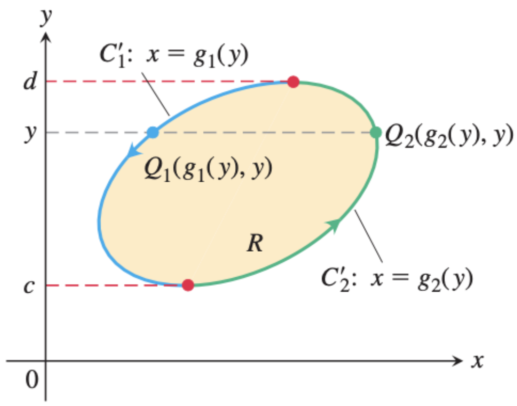
Now integrate both sides with respect to x from a to b

$$\begin{aligned}
\int_a^b \int_{f_1(x)}^{f_2(x)} \frac{\partial M}{\partial y} dy dx &= \int_a^b \left[M(x, f_2(x)) - M(x, f_1(x))\right] dx \\
&= -\int_b^a M(x, f_2(x))dx - \int_a^b M(x, f_1(x))dx \\
&= -\int_{C_2} M dx - \int_{C_1} M dx \\
&= -\oint_C M dx
\end{aligned}$$

Rewriting and reversing the order of integration

$$\oint_C M dx = \iint_R -\frac{\partial M}{\partial y} dx dy \tag{1}$$

Equation (1) is half the result to prove the circulation form of the Green's Theorem. We derive the other half by considering



Here C is made up of two directed paths:

$$\begin{aligned}
C_1' &:\rightarrow x = g_1(y), \quad c \leq y \leq d \\
C_2' &:\rightarrow x = g_2(y), \quad c \leq y \leq d
\end{aligned}$$

Mathematical Treatment: For any y between c and d, we can integrate $\dfrac{\partial N}{\partial x}$ with respect to x from $x = g_1(y)$ to $x = g_2(y)$

and obtain

$$
\begin{aligned}
\int_{g_1(y)}^{g_2(y)} \frac{\partial N}{\partial x} dx &= N(x,y)\Big|_{g_1(y)}^{g_2(y)} \\
&= N(g_2(y),y) - N(g_1(y),y)
\end{aligned}
$$

Now integrate both sides with respect to y from c to d

$$
\begin{aligned}
\int_c^d \int_{g_1(y)}^{g_2(y)} \frac{\partial N}{\partial x} dx dy &= \int_c^d \left[N(g_2(y),y) - N(g_1(y),y)\right] dy \\
&= \int_c^d N(g_2(y),y)dy - \int_c^d N(g_1(y),y)dy \\
&= \int_c^d N(g_2(y),y)dy + \int_d^c N(g_1(y),y)dy \\
&= \int_{C_2} N dy + \int_{C_1} N dy \\
&= \oint_C N dy
\end{aligned}
$$

Rewriting and reversing the order of integration

$$
\oint_C N dy = \iint_R \frac{\partial N}{\partial x} dx dy \tag{2}
$$

Summing (1) and (2) gives the equation

$$
\boxed{\oint_C \mathbf{F} \cdot \mathbf{T} \; ds = \oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy}
$$

Important Remark

Green's theorem, as stated, applies only to regions that are simply connected—that is, Green's theorem as stated so far cannot handle regions with holes. Here, we extend Green's theorem so that it does work on regions with finitely many holes.

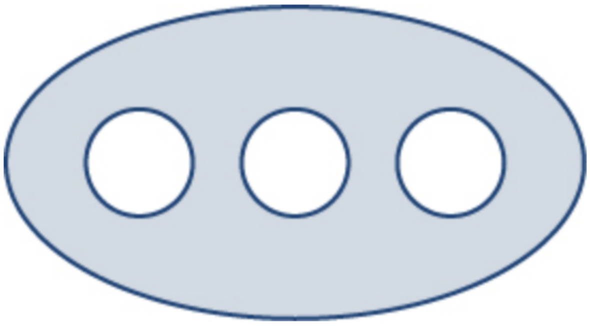


Figure 1: Green's Theorem, as stated, does not apply to a non simple connected region with three holes like this

The solution to this problem is to design the region in the following way

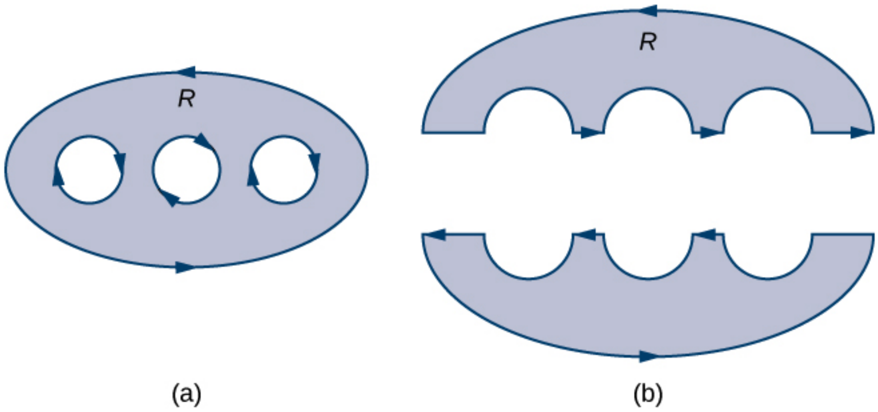


Figure 2: (a) Region D with an oriented boundary has three holes. (b) Region D is split into two simply connected regions with no holes.

Another example of doing this is

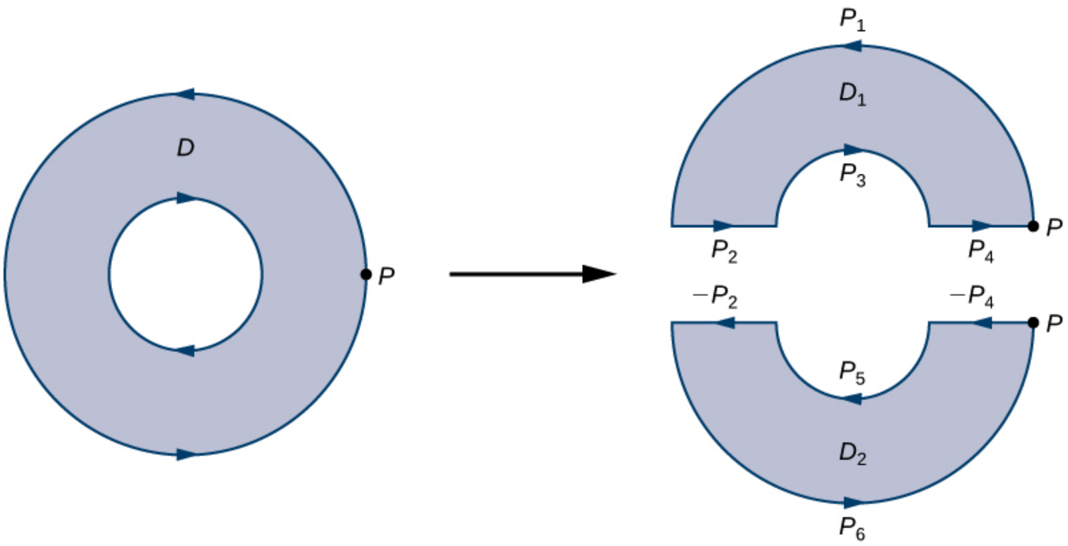


Figure 3: Breaking the annulus into two separate regions gives us two simply connected regions. The line integrals over the common boundaries cancel out.

The boundary of the upper half of the annulus, therefore, is $P_1 \cup P_2 \cup P_3 \cup P_4$ and the boundary of the lower half of the annulus is $-P_4 \cup P_5 \cup -P_2 \cup P_6$. Then, Green's Theorem implies

$$\begin{aligned}\int_{\partial D} \mathbf{F} \cdot d\mathbf{r} = &\int_{P_1} \mathbf{F} \cdot d\mathbf{r} + \int_{P_2} \mathbf{F} \cdot d\mathbf{r} + \int_{P_3} \mathbf{F} \cdot d\mathbf{r} \\ &+ \int_{P_4} \mathbf{F} \cdot d\mathbf{r} + \int_{-P_4} \mathbf{F} \cdot d\mathbf{r} + \int_{P_5} \mathbf{F} \cdot d\mathbf{r} + \int_{-P_2} \mathbf{F} \cdot d\mathbf{r} \\ &+ \int_{P_6} \mathbf{F} \cdot d\mathbf{r}\end{aligned}$$

Simplifying

$$
\begin{aligned}
\int_{\partial D} \mathbf{F} \cdot d\mathbf{r} &= \int_{P_1} \mathbf{F} \cdot d\mathbf{r} + \cancel{\int_{P_2} \mathbf{F} \cdot d\mathbf{r}} + \int_{P_3} \mathbf{F} \cdot d\mathbf{r} \\
&\quad + \cancel{\int_{P_4} \mathbf{F} \cdot d\mathbf{r}} - \cancel{\int_{P_4} \mathbf{F} \cdot d\mathbf{r}} + \int_{P_5} \mathbf{F} \cdot d\mathbf{r} \\
&\quad - \cancel{\int_{P_2} \mathbf{F} \cdot d\mathbf{r}} + \int_{P_6} \mathbf{F} \cdot d\mathbf{r} \\
&= \int_{P_1} \mathbf{F} \cdot d\mathbf{r} + \int_{P_3} \mathbf{F} \cdot d\mathbf{r} + \int_{P_5} \mathbf{F} \cdot d\mathbf{r} + \int_{P_6} \mathbf{F} \cdot d\mathbf{r} \\
&= \int_{\partial D_1} \mathbf{F} \cdot d\mathbf{r} + \int_{\partial D_2} \mathbf{F} \cdot d\mathbf{r} \\
&= \iint_{D_1} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA + \iint_{D_2} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA \\
&= \iint_{D} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA
\end{aligned}
$$

Examples

- Verify Green's Theorem for the line integral along the unit circle C, oriented counterclockwise

$$\oint_C xy^2 dx + x dy$$

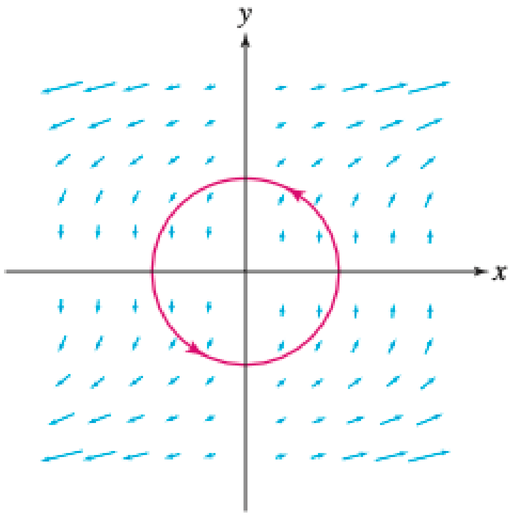


Figure 4: The vector field $\mathbf{F}(x, y) = \langle xy^2, x \rangle$

- Compute the circulation of $\mathbf{F}(x, y) = \langle \sin x, x^2 y^3 \rangle$ around the triangular path C with the counterclockwise orientiation shown as

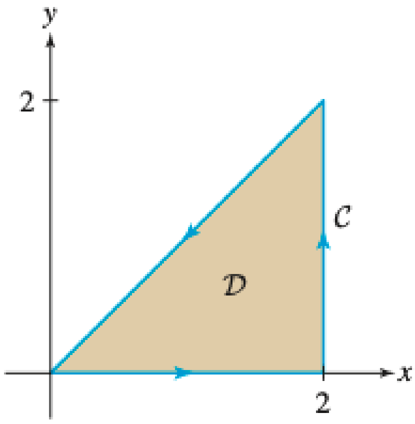


Figure 5: The region D is described by $0 \le x \le 2$, $0 \le y \le x$

- **Using Green's Theorem on a Region with Holes**

Calculate the integral

$$\oint_C \left(\sin(x) - \frac{y^3}{3}\right) dx + \left(\frac{x^3}{3} + \sin(y)\right) dy$$

where C is the annulus given by the polar inequalities $1 \leq r \leq 2$, $0 \leq \theta \leq 2\pi$

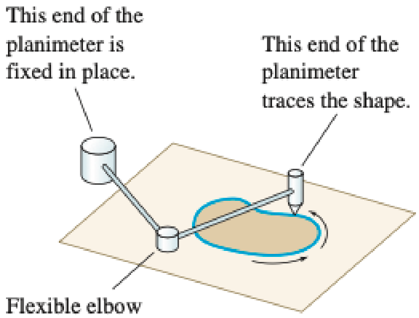
Area via Green's Theorem

If a simple closed curve C in the plane and the region R it encloses satisfy the hypothesis of Green's Theorem, the area of R is given by

$$\text{Area of R} = \oint_C xdy = \oint_C -ydx = \frac{1}{2}\oint_C xdy - ydx$$

Hint: To prove, consider the special case of vector fields $\mathbf{F} = \langle 0, x\rangle$ or $\mathbf{F} = \langle -y, 0\rangle$ or $\mathbf{F} = \langle -\frac{y}{2}, \frac{x}{2}\rangle$

These remarkable formulas tell us how to compute an enclosed area by making measurements only along the boundary. It is the mathematical basis of the **planimeter**, a device that computes the area of an irregular shape when you trace the boundary with a pointer at the end of a movable arm.



A cool animation of Linear Planimeter in action.

Area via Green's Theorem

Compute the area of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

using line integral.

Using the Extended Form of Green's Theorem

Let $\mathbf{F} = \langle M, N \rangle = \left\langle \frac{y}{x^2+y^2}, -\frac{x}{x^2+y^2} \right\rangle$ and C be any simple closed curve in a plane oriented counterclockwise. What are the possible values of $\int_C \mathbf{F} \cdot d\mathbf{r}$?

We start by considering two cases: the case when C encompasses the origin and the case when C does not encompass the origin.

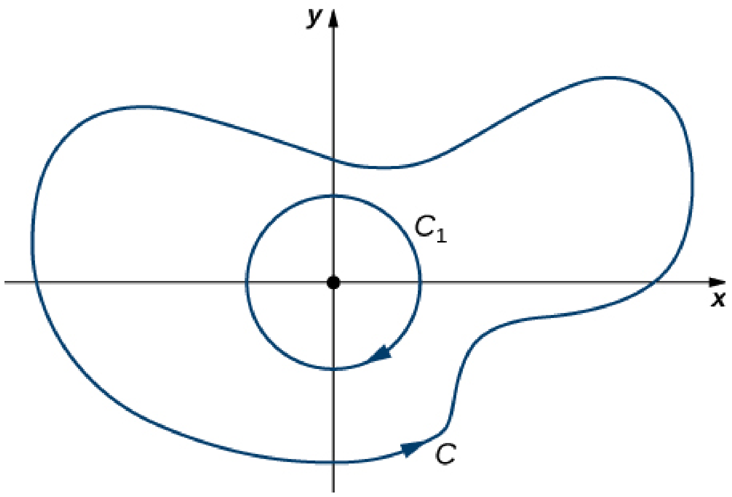
Case-1: Does Not Encompass the Origin

In this case, the region enclosed by C is simply connected because the only hole in the domain of $\mathbf{F}$ is at the origin. $\mathbf{F}$ satisfies the cross partial condition on the restricted domain, this implies $\mathbf{F}$ is conservative on this simple connected region and hence

$$\int_C \mathbf{F} \cdot d\mathbf{r} = 0$$

Case-1: Does Encompass the Origin

In this case, the region enclosed by C is not simply connected because the region contains a hole at the origin. Let C_1 be a circle of radius a centered at origin so that C_1 is entirely inside the region enclosed by C. Give C_1 a clockwise orientation.



Let D be the region between C_1 and C, and is oriented counter-clockwise. By the extended version of Green's Theorem,

$$\begin{aligned}\int_C \mathbf{F}\cdot d\mathbf{r} + \int_{C_1} \mathbf{F}\cdot d\mathbf{r} &= \iint_D (N_x - M_y)dA \\ &= \iint_D \left(-\frac{y^2-x^2}{(x^2+y^2)^2} + \frac{y^2-x^2}{(x^2+y^2)^2}\right) dA \\ &= 0\end{aligned}$$

This implies that

$$\int_C \mathbf{F}\cdot d\mathbf{r} = -\int_{C_1} \mathbf{F}\cdot d\mathbf{r}$$

and we can evaluate the integral on the right side by using the parameterization

$$C_1 : \Longrightarrow \quad x = a\cos(t), y = a\sin(t),\ 0 \le t \le 2\pi$$

Therefore,

$$\int_{C_1} \mathbf{F}\cdot d\mathbf{r} = 2\pi$$

and

$$\int_C \mathbf{F}\cdot d\mathbf{r} = -2\pi$$